<u>Titanic Survival Prediction</u>

Name: Kisothana Palasuntharam

Internship: DecodeLabs Data Science Internship

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1. Introduction

The objective of this project is to analyze the Titanic dataset and predict passenger survival using Machine Learning.

2. Dataset Description

The dataset contains information about Titanic passengers such as age, gender, passenger class, fare, and survival status.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

3. Data Cleaning

- Checked missing values

- Handled null data

- Verified data types

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

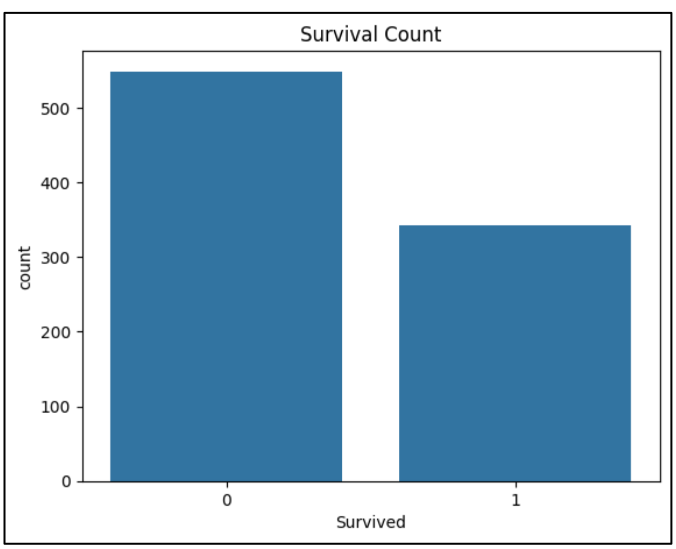
4. Exploratory Data Analysis

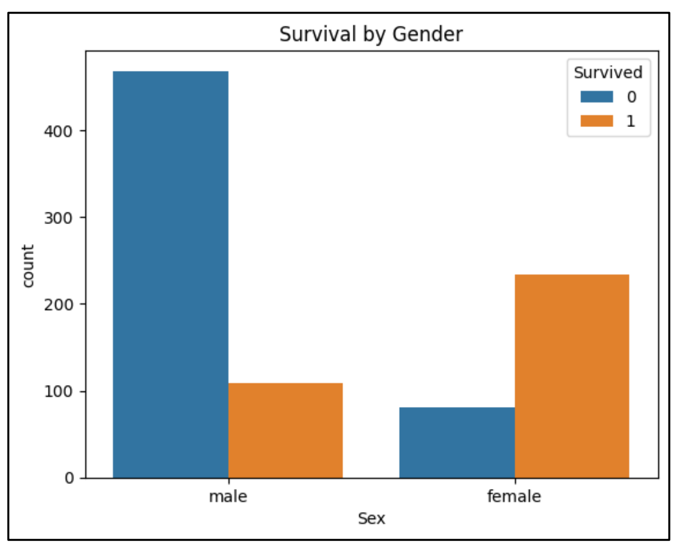
- Survival distribution

- Gender-wise survival analysis

- Passenger class analysis

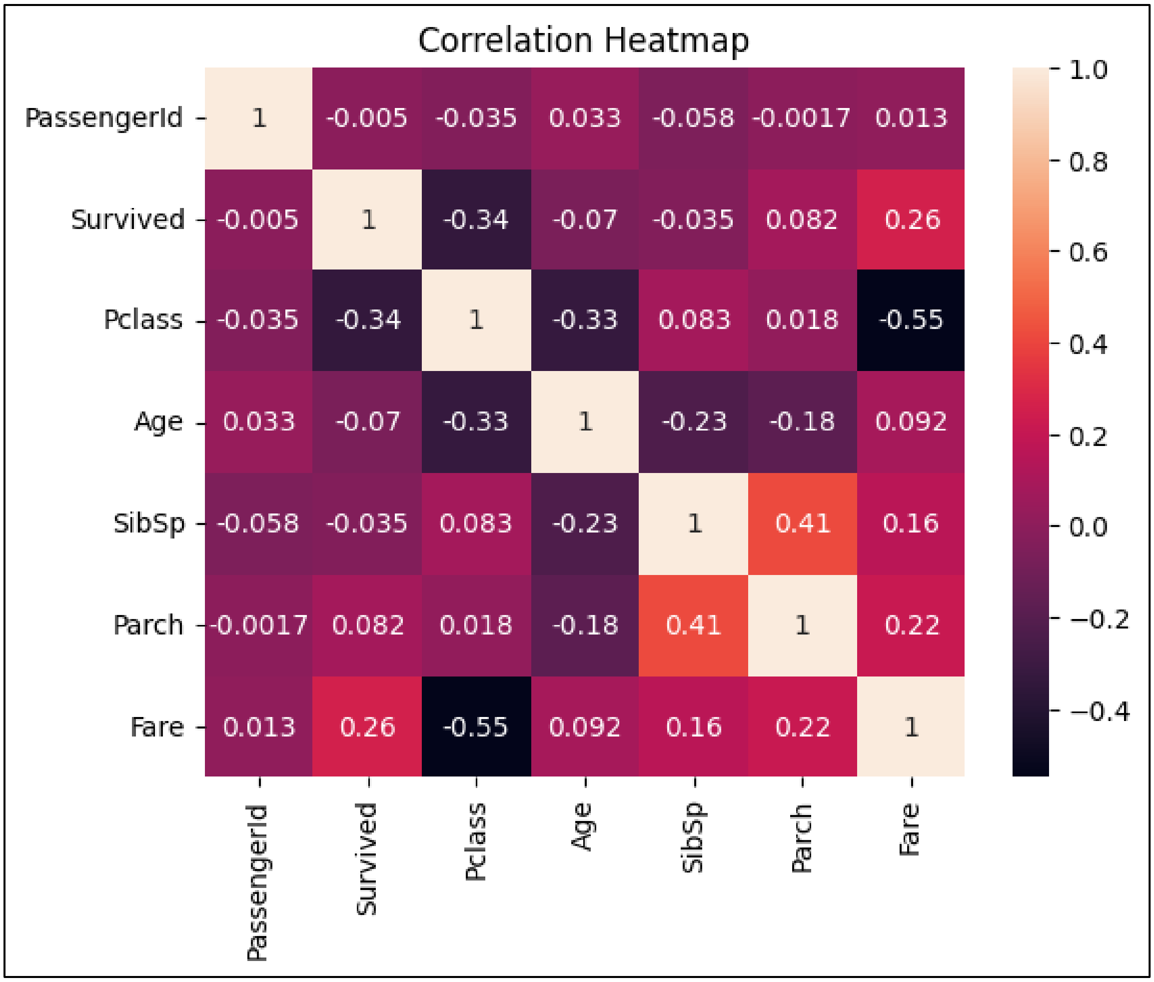
- Age distribution





5. Data Visualization

Created charts using Matplotlib and Seaborn to understand patterns in the data.



6. Machine Learning Model

Model Used: Logistic Regression

Features:

- Pclass
- Sex
- Age
- Fare

Target:

- Survived

```python
# Select Important Features
features = ['Pclass', 'Sex', 'Age', 'Fare']

# Convert Text to Numbers
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# Define X and y
X = df[features]

y = df['Survived']

# Split Training & Testing Data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create ML Model
model = LogisticRegression()
```

7. Results

The model successfully predicted passenger survival and demonstrated the complete machine learning workflow.

```python
# Check Accuracy
accuracy = accuracy_score(y_test, predictions)

print("Logistic Regression Accuracy:", accuracy)
```

```
Logistic Regression Accuracy: 0.7988826815642458
```

```python
# Test Custom Passenger Prediction
sample_passenger = [[1, 1, 25, 100]]

prediction = model.predict(sample_passenger)

print(prediction)
```

```
[1]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not h
  warnings.warn(
```

8. Conclusion

This project helped develop practical skills in data cleaning, visualization, exploratory data analysis, and machine learning.